

opamp_poly

GF180MCU Precision Operational Amplifier

GlobalFoundries 180 nm MCU PDK

Rev. 1.0 June 2026

80.3 dB

Open-Loop Gain

154.6 MHz

Unity-GBW

70.2°

Phase Margin

102.8 dB

CMRR

194 V/ μ s

Slew Rate

1 General Description

The opamp_poly is a high-performance, internally compensated two-stage Miller-compensated operational amplifier implemented in the GlobalFoundries 180 nm MCU process (GF180MCU PDK, gf180mcuD). It is designed for ultra-low-noise, high-precision analog front-ends, featuring an exceptionally wide unity-gain bandwidth, high open-loop gain, and excellent common-mode and power-supply rejection.

Primary applications include transimpedance amplifiers (TIA) for electrochemical gas-sensor ASICs, precision level-shifting gain stages, and any mixed-signal sensor interface requiring a stable, well-compensated amplifier core.

Key Features

- **80.3 dB** open-loop gain
- **154.6 MHz** unity-gain bandwidth
- **70.2°** phase margin (internally compensated)
- **102.8 dB** CMRR / **81.9 dB** PSRR
- Rail-to-rail output swing: 0 V – 3.23 V @ $V_{DD} = 3.3$ V
- Single **3.3 V** supply; $I_Q = 943 \mu\text{A}$
- Wide ICMR: 38 mV – 3.06 V

Applications

- Electrochemical sensor TIA / potentiostat front-end
- MOX sensor signal conditioning
- Precision virtual-ground generators
- Active filter stages in industrial ASICs

2 Pin Configuration

Pin	Type	Description
vdd	Pwr	Positive supply (typ. 3.3 V)
gnd	Pwr	Ground reference (0 V)
vp	In	Non-inverting input
vn	In	Inverting input
vbias	In	Bias voltage for tail current mirror (typ. 1.0 V)
out	Out	Amplifier output

3 Absolute Maximum Ratings

Exceeding these limits may cause permanent damage.

Parameter	Sym.	Rating
Supply Voltage	V_{DD}	–0.3 to 3.6 V
Input Voltage	V_{IN}	0 to V_{DD}
Operating Temp.	T_A	–40 to 125 °C

4 Electrical Characteristics

Unless otherwise noted: $V_{DD} = 3.3$ V, $V_{bias} = 1.0$ V, $V_{CM} = 1.3$ V, $T_A = 27$ °C, $C_{load} = 25$ fF.

DC Performance

Parameter	Sym.	Min	Typ	Max
Open-Loop Gain	A_{VOL}	–	80.3 dB	–
Input Common-Mode Range	V_{ICMR}	38 mV	–	3.06 V
CMRR	CMRR	–	102.8 dB	–
PSRR	PSRR	–	81.9 dB	–
Output High	V_{OH}	–	3.23 V	–
Output Low	V_{OL}	–	≈0 V	–

AC Performance & Dynamics

Parameter	Sym.	Min	Typ	Max
Unity-Gain BW	UGBW	–	154.6 MHz	–
Phase Margin	PM	–	70.2°	–
Slew Rate (+)	SR_+	–	194.4 V/ μ s	–
Slew Rate (–)	SR_-	–	96.0 V/ μ s	–

Power Supply

Parameter	Sym.	Min	Typ	Max
Supply Voltage	V_{DD}	3.0 V	3.3 V	3.6 V
Quiescent Current	I_Q	–	943 μ A	–
Static Power	P_D	–	3.11 mW	–

5 Typical Performance Characteristics

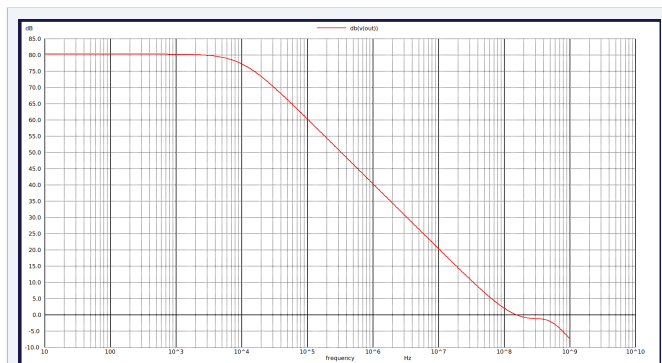


Figure 1: Simulated AC open-loop gain and phase response ($V_{DD} = 3.3\text{ V}$, $V_{bias} = 1.0\text{ V}$, $C_{load} = 25\text{ fF}$).

6 Design Notes

Compensation. The `opamp_poly` uses Miller compensation (C_C , R_C network between first and second stage) to achieve a dominant pole and the specified 70.2° phase margin at the rated load.

Bias Pin (v_{bias}). An external 1.0 V reference sets the tail current of the input differential pair via an internal mirror. Deviating significantly from 1.0 V will alter I_Q , GBW, and SR proportionally.

Slew Rate Asymmetry. The rising SR ($194\text{ V}/\mu\text{s}$) exceeds the falling SR ($96\text{ V}/\mu\text{s}$) due to the asymmetric sourcing/sinking drive capability of the output stage. Account for this in settling-time budgets for the TIA application.

7 Integration & Ordering Information

Field	Value
Macro name	<code>opamp_poly</code>
PDK	GF180MCU (<code>gf180mcuD</code>)
EDA tools	Xschem + NGspice
Supply	3.3 V (single)
Team	NanoArchitects B15
Contact	A. Karampurwala, VIT Vellore

Simulation results obtained with NGspice using `gf180mcuD` SPICE models at TT corner, 27°C . All specifications subject to change pending post-layout simulation and silicon characterisation.